

## Level Mudah

### Tugas 1 – Membuat Node Sederhana dan Menampilkan Datanya

```
class Node:
    def __init__(self, data):
        self.data = data

node1 = Node(10)

print("Data Node:", node1.data)
```

Data Node: 10

### Tugas 2 – Membuat Linked List Berisi 5 Angka

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

# Membuat node
node1 = Node(10)
node2 = Node(20)
node3 = Node(30)
node4 = Node(40)
node5 = Node(50)

# Menghubungkan node
node1.next = node2
node2.next = node3
node3.next = node4
node4.next = node5

head = node1
```

### Tugas 3 – Traversal Seluruh Node

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

node1 = Node(10)
node2 = Node(20)
node3 = Node(30)
node4 = Node(40)
node5 = Node(50)

node1.next = node2
node2.next = node3
node3.next = node4
node4.next = node5

current = node1

print("Isi Linked List:")
while current:
    print(current.data, end=" -> ")
    current = current.next

print("NULL")
```

Isi Linked List:  
10 -> 20 -> 30 -> 40 -> 50 -> NULL

### Tugas 4 – Menambahkan Node di Awal Linked List

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

# Linked List awal
head = Node(10)
```

```

head.next = Node(20)
head.next.next = Node(30)

# Insert di awal
new_node = Node(5)
new_node.next = head
head = new_node

# Tampilkan
current = head
while current:
    print(current.data, end=" -> ")
    current = current.next

print("NULL")

```

```
5 -> 10 -> 20 -> 30 -> NULL
```

## Level Menengah

### Tugas 5 – Menambahkan Node di Akhir Linked List

```

class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

head = Node(10)
head.next = Node(20)
head.next.next = Node(30)

# Node baru
new_node = Node(40)

current = head
while current.next:
    current = current.next

current.next = new_node

# Tampilkan
current = head
while current:
    print(current.data, end=" -> ")
    current = current.next

print("NULL")

```

```
10 -> 20 -> 30 -> 40 -> NULL
```

### Tugas 6 – Menghapus Node Tertentu

Misalnya menghapus data 20

```

class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

head = Node(10)
head.next = Node(20)
head.next.next = Node(30)
head.next.next.next = Node(40)

hapus = 20

current = head

while current.next:
    if current.next.data == hapus:
        current.next = current.next.next
        break
    current = current.next

# Tampilkan hasil
current = head
while current:
    print(current.data, end=" -> ")
    current = current.next

```

```
10 -> 30 -> 40 -> NULL
```

### Tugas 7 – Mencari Data pada Linked List

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

head = Node(10)
head.next = Node(20)
head.next.next = Node(30)
head.next.next.next = Node(40)

cari = 30

current = head
ditemukan = False

while current:
    if current.data == cari:
        ditemukan = True
        break
    current = current.next

if ditemukan:
    print("Data ditemukan")
else:
    print("Data tidak ditemukan")
```

Data ditemukan

### Tugas 8 – Menghitung Jumlah Node

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

head = Node(10)
head.next = Node(20)
head.next.next = Node(30)
head.next.next.next = Node(40)
head.next.next.next.next = Node(50)

jumlah = 0
current = head

while current:
    jumlah += 1
    current = current.next

print("Jumlah Node:", jumlah)
```

Jumlah Node: 5

